

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
M2: Enterprise Java

Practical-9

Roll No.:T017	Name: Radhika mali
Class:TYIT	Batch:1
Date of Assignment: 12-9-2026	Date/Time of Submission: 12-9-2026

Aim:- Implement the following JPA applications.

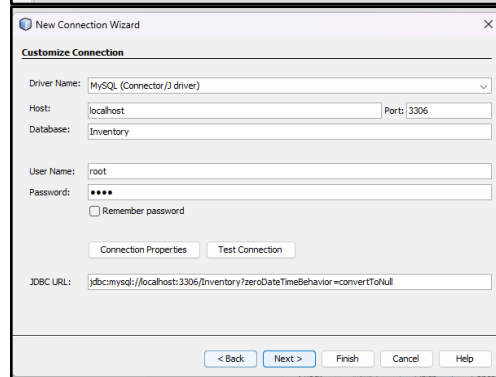
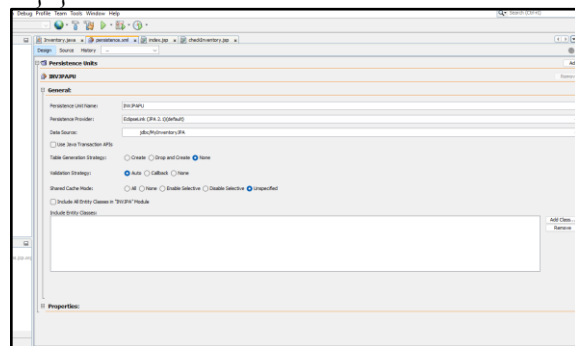
A. Develop a simple Inventory Application Using JPA.

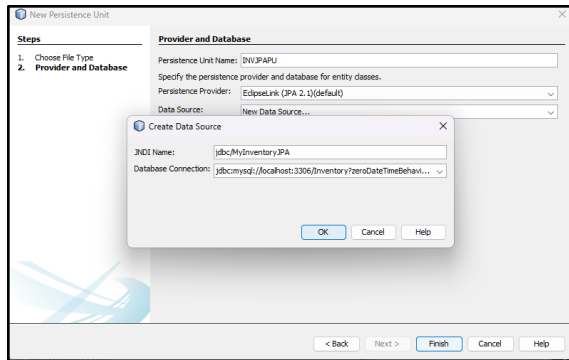
Code:-

Inventory.java

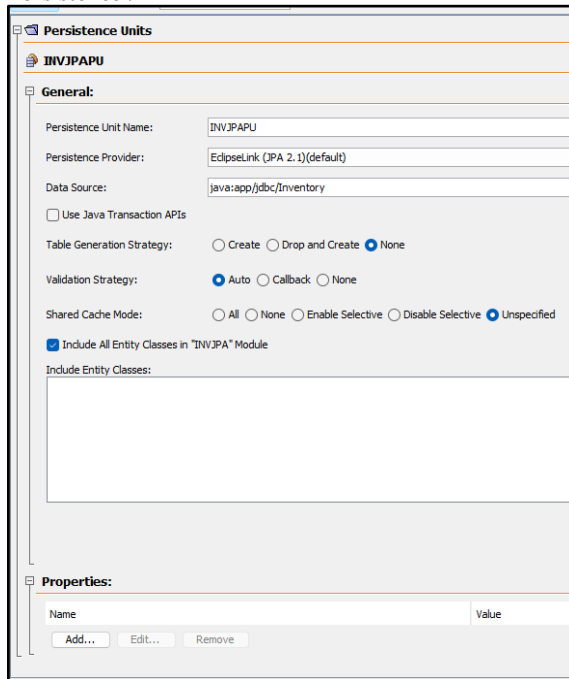
```
package myApp;
import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;
@Entity
@Table(name = "inventory")
public class Inventory {
    @Id
    @GeneratedValue(strategy =
    GenerationType.IDENTITY)
    @Column(name = "itemid", unique = true, updatable =
    false)
    private Integer itemid;
    @Column(name = "itemName")
    private String itemName;
    @Column(name = "description")
    private String description;
    @Column(name = "quantity")
    private Integer quantity;
    @Column(name = "addedDate")
    private String addedDate;
    public Inventory() {
    }
    public Integer getItemID() {
        return itemid;
    }
    public void setItemID(Integer itemid) {
        this.itemid = itemid;
    }
    public String getItemName() {
        return itemName;
    }
    public void setItemName(String itemName) {
        this.itemName = itemName;
    }
    public String getDescription() {
        return description;
    }
}
```

```
public void setDescription(String description) {
    this.description = description;
}
public Integer getQuantity() {
    return quantity;
}
public void setQuantity(Integer quantity) {
    this.quantity = quantity;
}
public String getAddedDate() {
    return addedDate;
}
public void setAddedDate(String addedDate) {
    this.addedDate = addedDate;
}
}
```





Persistence.xml



Sql :

```
mysql> CREATE DATABASE Inventory;
Query OK, 1 row affected (0.01 sec)

mysql> USE Inventory;
Database changed
mysql>
mysql> CREATE TABLE inventory (
  ->   itemid INT PRIMARY KEY AUTO_INCREMENT,
  ->   itemName VARCHAR(50),
  ->   description VARCHAR(200),
  ->   quantity INT,
  ->   addedDate VARCHAR(40)
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql> USE Inventory;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_inventory |
+-----+
| inventory            |
+-----+
1 row in set (0.03 sec)
```

Index.jsp:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
```

```
<!DOCTYPE html>
<html>
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
  <title>Inventory</title>
</head>
<body>
<form action="checkInventory.jsp" method="post">
<table>
<tr>
  <td colspan="2" align="center">
    <b>Add the item in Inventory</b>
  </td>
</tr>
<tr>
  <td>Item Name:</td>
  <td>
    <input name="itemname" maxlength="25" size="50"/>
  </td>
</tr>
<tr>
  <td>Description:</td>
  <td>
    <textarea rows="5" cols="36" name="description"></textarea>
  </td>
</tr>
<tr>
  <td>Quantity:</td>
  <td>
    <input name="quantity" maxlength="4" size="5"/>
  </td>
</tr>
<tr>
  <td colspan="2">
    <input type="submit" name="AddToInventory" value="AddToInventory"/>
  </td>
</tr>
</table>
</form>
</body>
</html>
```

checkInventory.jsp:

```

<%@page import="java.util.Iterator"%>
<%@page import="javax.persistence.Persistence"%>
<%@page import="java.util.List"%>
<%@page import="myApp.Inventory"%>
<%@page
import="javax.persistence.EntityTransaction"%>
<%@page import="javax.persistence.EntityManager"%>
<%@page
import="javax.persistence.EntityManagerFactory"%>
<%@page contentType="text/html" pageEncoding="UTF-
8"%>
<%!
    private EntityManagerFactory eMF;
    private EntityManager eM;
    private EntityTransaction eT;
    List<Inventory> inv;
%>
<%
    eMF =
Persistence.createEntityManagerFactory("INVJPAPU");
    eM = eMF.createEntityManager();
    String submit =
request.getParameter("AddToInventory");
    if (submit != null && submit.equals("AddToInventory"))
    {
        try {
            String itemname =
request.getParameter("itemname");
            String description =
request.getParameter("description");
            String quantity1 = request.getParameter("quantity");
            int quantity = Integer.parseInt(quantity1);
            String addedDate = new java.util.Date().toString();
            Inventory inventory = new Inventory();
            inventory.setItemName(itemname);
            inventory.setDescription(description);
            inventory.setQuantity(quantity);
            inventory.setAddedDate(addedDate);
            eT = eM.getTransaction();
            eT.begin();
            eM.persist(inventory);
            eT.commit();
        } catch (RuntimeException e) {
            if (eT != null && eT.isActive()) {
                eT.rollback();
            }
            throw e;
        }
        response.sendRedirect("checkInventory.jsp");
        return;
    }
    inv = eM.createQuery(
        "SELECT i FROM Inventory i",
        Inventory.class
    ).getResultList();
    eM.close();
    eMF.close();
%>

```

```

<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Inventory</title>
</head>
<body>
<table border="1">
<tr>
    <td colspan="5" align="center">
        <b>
            Click <a href="index.jsp">here</a>
            to add items in the Inventory.
        </b>
    </td>
</tr>
<tr>
    <td>Item No</td>
    <td>Item Name</td>
    <td>Description</td>
    <td>Quantity</td>
    <td>Added Date & Time</td>
</tr>
<%
    Iterator<Inventory> iterator = inv.iterator();
    while (iterator.hasNext()) {
        Inventory objInv = iterator.next();
%>
<tr>
    <td><b><%=objInv.getItemID()%></b></td>
    <td><b><%=objInv.getItemName()%></b></td>
    <td><b><%=objInv.getDescription()%></b></td>
    <td><b><%=objInv.getQuantity()%></b></td>
    <td><b><%=objInv.getAddedDate()%></b></td>
</tr>
<%
    }
%>
</table>
</body>
</html>

```

Output:-

Add the item in Inventory

Item Name:

Description:

Quantity:

Click here to add items in the Inventory.				
Item No	Item Name	Description	Quantity	Added Date & Time
1	Mobile	Samsung Galaxy Note 2	30	Sat Sep 12 10:28:56 IST 2026